Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

Anonymous Author(s)

ABSTRACT

Quoted stablecoin price gaps are not trades. We introduce Stablecoin StressBench, a reproducible benchmark that separates optical dislocations from executable arbitrage opportunities using per-minute order-book simulation. During the March 2023 USDC/SVB stress event, 34.3% of minutes exceed a 10 bps price basis; only 2.88% remain net-profitable after execution costs — a 12× gap. No model trained on calm-market data earns positive returns on executable tasks; the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% of oracle returns), demonstrating that execution microstructure is mechanism-invariant. A PPO-GRU agent trained on the same cross-mechanism protocol converges to NoTrade: the 2.3% positive rate in the Terra/LUNA training split produces a reward signal too sparse to calibrate entry timing, confirming that explicit positive examples at training time — not sequential observability — are the key requirement. To our knowledge, this is the first benchmark to provide execution-aware stablecoin labels with route-depth provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:
Anonymous Author(s). 2026. Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations. In Proceedings of ACM International Conference on AI in Finance (ICAIF '26). ACM, New York, NY, USA, 8 pages.

1 INTRODUCTION

Stablecoins are dollar-pegged cryptocurrencies that underpin the majority of DeFi lending, trading, and payments. During stress events, peg fragmentation creates cross-venue price gaps that look like arbitrage opportunities: buy cheaply on one exchange, sell at par on another, capture the spread. In practice, however, a quoted price gap is not a trade. Clearing the required order-book depth on a large notional moves the execution price; VWAP slippage, taker fees, and CEX-to-CEX settlement latency each add frictions that are small in isolation but jointly eliminate most of the apparent profit. The result is that many stablecoin dislocations are illusory: they appear profitable on a price chart but disappear once the cost of executing a real order through a live order book is accounted for. We call these optical dislocations, to distinguish them from executable ones where net profit survives all transaction costs.

Prior machine-learning work on stablecoin predictability uses price-based labels to define "profitable" windows, which conflates the optical and executable layers and systematically overstates how predictable actual trading opportunities are. Stablecoin StressBench is designed to close this gap. It provides per-minute execution-aware labels that separate three dislocation layers: (1) an optical dislocation (quoted basis exceeds a threshold), (2) an executable dislocation (net profit is positive after VWAP execution, fees, and settlement at a specified notional), and (3) a predictable dislocation (a model correctly identifies the window ex ante). The distance between these layers is the paper's core measurement.

We build the benchmark on the March 2023 USDC/SVB stress event, where Silicon Valley Bank's collapse triggered a brief USDC depeg. Using real Binance L2 order-book data and a full VWAP book-walking model, we find that 34.3% of 1-minute windows show a > 10 bps price basis but only 2.88% remain net-profitable at \$10K notional: a 12× price-to-execution gap. We then construct a model ladder spanning rule-based thresholds, statistical baselines, gradient-boosted classifiers, sequence models (GRU), and meta-labeling filters, evaluating each model's ability to identify the 2.88% of executable windows ex ante. Every model trained on calm-control data loses money on executable arbitrage tasks; the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling, trained on a different stress event (Terra/LUNA 2022), is the only approach that achieves positive net returns (+82.5 bps, 50.8% oracle capture).

Contributions.

(1) Execution-aware benchmark: per-minute labels with documented order-book depth provenance, VWAP book-walking, fees, notional scaling, and a hindsight oracle ceiling.
(2) 18-event stress taxonomy: seven mechanism classes (2020–2023) with data-tier governance of permitted claims. Terra/LUNA cross-mechanism check gives 5.9× vs. the Tier-A 12× ratio.
(3) Oracle gap evidence: +162 bps oracle vs. +17.2 bps best model (basis task), > 260 bps gap on executable arbitrage (structural, not label-mismatch).
(4) Cross-mechanism transfer and mechanism-invariant microstructure: A meta-model trained on Terra/LUNA dynamics and evaluated on USDC/SVB achieves +82.5 bps net (50.8% oracle capture), demonstrating that order-book depth features predict executable windows across distinct stress mechanisms. Oracle-gap decomposition

ICAIF '26, November 14–17, 2026, Milan, Italy
2026.

identifies false-positive cost and timing as the dominant structural barriers.

Paper organisation. Section 2 reviews related work. Section 3 describes benchmark design and data. Section 4 defines execution-aware labels. Section 5 describes the model ladder and evaluation. Section 6 presents all results, including the oracle gap, cross-mechanism transfer, SHAP analysis, oracle gap decomposition, and RL execution policy evaluation. Section 7 discusses limitations. Section 8 concludes.

2 RELATED WORK

Stablecoin stress. The USDC/SVB episode produced well-documented price fragmentation; we address ex-ante predictability. Uhlig [1] analyses Terra/LUNA mechanics (our validation split); Griffin and Shams [2] study Tether manipulation (our focus is execution microstructure).

Limits to arbitrage and crypto fragmentation. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish theory. Makarov and Schoar [7] show persistent cross-venue crypto price gaps that are not executable profits. Hautsch et al. [3] show settlement latency is a first-order arbitrage cost; Cont and Kukanov [6] show VWAP price (not quoted price) is the relevant object.

Financial ML and benchmarks. López de Prado [8] introduces meta-labeling; Adams and MacKay [15] introduce BOCPD. Moallemi and Wang [16] and Ning et al. [17] apply RL to optimal order execution in equity markets; we apply the same framing to stress-event arbitrage timing in crypto, where the training-distribution mismatch identified in §6.4 makes cross-mechanism initialisation a necessary precondition. Cintra and Holloway [9] detect the Terra depeg 12 hours early via Curve AMM pool reserves. We find BOCPD achieves only AUROC 0.23 on CEX cross-quote data, because AMM reserves accumulate gradually while CEX basis is mean-reverting, making changepoint detection unreliable on quoted prices. Stablecoin StressBench is, to our knowledge, the first benchmark to provide per-minute execution-aware labels with documented order-book depth provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation across distinct stress mechanisms.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: executable rate falls 2.88%→2.40%, ratio widens 12×→14×, and the oracle stays above +130 bps: the headline result is robust to buy-leg proxy error.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits. The 2.88% positive rate implies severe class imbalance; AUROC is reported but economic net bps is the primary criterion.

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

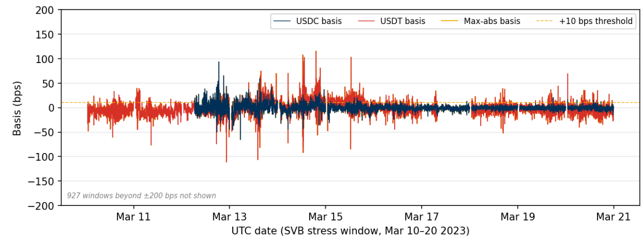

Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023; y-axis clipped to ±200 bps; 927 windows with $|b|>200$ bps not shown, all USDC-side spikes during peak reserve uncertainty on Mar 10–12). USDC basis (navy): 12.4% of minutes > 10 bps. USDT basis (red): 27.4%. Max-abs (gold): 34.3%. Large optical dislocations, but the price-to-execution gap is 12×.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure 1). No model trained on the train split has seen stress dynamics; threshold calibration on validation uses an independent stress event. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit checks confirming no future information leaks into the input features.

3.3 Historical Stress Taxonomy and Scope

We catalogue 18 stress events (2020–2023) across seven mechanism classes: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD); fiat-reserve bank shock (USDC/SVB); regulatory winddown (BUSD); exchange credit (Celsius/3AC, HUSD, FTX [10]); DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI); collateral liquidation (DAI Black Thursday); RWA/niche (Acala aUSD, USDR).

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A (N=2)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B (N=11)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C (N=5)	RWA, niche	Thin market	Taxonomy only

Data availability governs permitted claims (Table 3). Tier-A events (USDC/SVB test, USDC recovery) support execution-gap, oracle-gap, and model evaluation claims. Tier-B (N=11, OHLCV/pool data) support price magnitude estimates only. Tier-C (N=5, partial/DEX-only) support mechanism taxonomy only.

3.4 Feature Sets

Four nested feature sets (Table 3.4) isolate marginal information value:

`price_only`	5 cross-quote basis cols
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

4 EXECUTION-AWARE LABELS

Labels operationalise the three-layer framework introduced in §1. The price-to-execution gap measures Layer 1 ÷ Layer 2; the oracle gap measures Layer 3 vs. Layer 2 in hindsight. This section details label construction.

4.1 Cross-Quote Basis and Labels

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad \text{[bps]}$$

$b_{\max} = \max(|b_{\mathrm{USDC}}|, |b_{\mathrm{USDT}}|)$. Basis label: $y_{\mathrm{basis}}(t) = \mathbf{1}[|b_{\mathrm{USDC}}(t{+}h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) = \mathrm{VWAP}_{\mathrm{sell}} - \mathrm{VWAP}_{\mathrm{buy}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}} \tag{1}$$

$$y_{\mathrm{arb}}(q,h,t) = \mathbf{1}\left[\max_{s\in[t,\,t+h]} \mathrm{net}(q,s) > c\right] \tag{2}$$

Parameters: c=10 bps, taker fee 4 bps/side, δ=5 bps. Primary economic task: `executable_arb_q10000_5m`. The oracle trades every minute where $\mathrm{net}(q,t) > c$ in hindsight, setting the performance ceiling.

5 MODELS AND EVALUATION

5.1 Model Ladder

The ladder tests whether stronger models close the oracle gap, with each rung testing a distinct hypothesis: threshold rules test whether price signals alone suffice; ML classifiers test whether microstructure conditioning helps; the expected-net-profit regressor tests whether the gap is a label-mismatch artefact; sequence models test whether temporal depth patterns are exploitable; cross-mechanism meta-labeling tests whether stress microstructure is mechanism-invariant.

Economic anchors. `NoTrade` (0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis{10,25}bps` and `GrossArbThreshold`: threshold rules on price data.

Statistical baselines. `LastValue`, `RollingMean`, `AR1`: basis autocorrelation only.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, LightGBM: trained on four nested feature sets.

Expected-net-profit regressor. Directly regresses $\hat{y} = \mathrm{net}(q,t)$ and trades when $\hat{y} > \theta^*$. If this fails, the gap is structural, not label-mismatch.

Meta-labeling filter. Primary signal: $|b_{\mathrm{USDC}}| > 10$ bps; secondary LightGBM meta-model learns $\Pr[\mathrm{net}(q,t) > c \mid \text{primary fires}]$.

Regime detectors. EWMA z-score, CUSUM, BOCPD as two-stage stress-regime pre-filters.

Sequence models. GRU and MLP-Window classifiers trained on 30-minute sliding windows of order-book microstructure features (§6.4).

5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \mathrm{net}(q,t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features — the smallest feature set delivering the dominant SHAP lift over price-only signals (§6.6) — available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol: training on Terra/LUNA stress windows only, evaluation on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.8.

5.3 Calibration and Evaluation

Threshold calibration on validation split: $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \mathrm{net}(q,t_i)$ subject to $\geq$ 25 trades. Primary metrics: `net_bps_captured` and `oracle_capture_pct` = model net bps ÷ oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary: AUROC, AUPRC. `NoTrade` at 0 bps is the economic floor.

Quantitative claim boundaries: (A1) BTC-USDC buy leg uses kline-proxy for SVB window; (A2) oracle-gap results from SVB test split (N=15,832) only; (A3) c=10 bps, 4 bps fee/side, δ=5 bps.

6 RESULTS

6.1 Price Gaps Are Common; Execution Is Rare

At 10 bps, 34.3% of test minutes exceed the max-basis threshold; only 2.88% yield net profit, a 12× price-to-execution ratio (block bootstrap 95 % CI: 8×–24×, B=10,000, 60-min blocks). The gap persists at all tested thresholds (Table 4)

Table 4: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

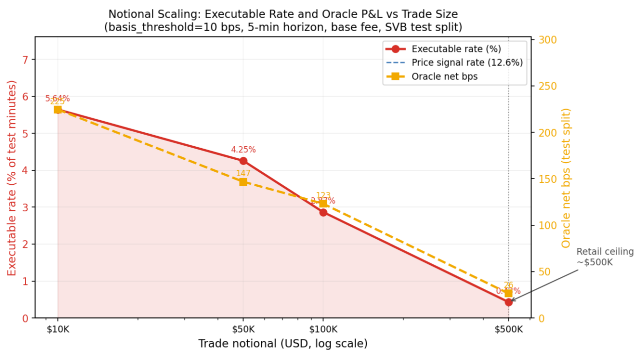

Figure 2: Executable rate (%, red, left) and oracle net bps (gold, right) vs. trade size (log scale). Sharp elbow at $10K–$50K marks the retail arbitrage ceiling: book depth eliminates the opportunity above this notional.

Table 5: Oracle gap summary (test split). *=only non-oracle tabular result above zero; ‡=real Tier-A L2 sequence model.

Task / Model	Oracle	Best model	Gap
`basis_usdc_1m`	+162.2 bps	+17.2 bps*	145 bps
`exec_arb_q10k_5m`	+225.0 bps	−41.1 bps	266 bps
`exec_arb_q50k_5m`	+147.1 bps	−113.0 bps	260 bps
GRU (seq)‡, `basis_usdc_1m`	+162.2 bps	−239.0 bps	401 bps

*`lgbm@price_book_settle` feature set, 106 trades; best arb: `price_threshold_25bps@price_only`.

‡GRU, 30-min window, real Tier-A L2; gap is structural even at AUROC 0.80 (§6.4).

and is robust across cost regimes: at the worst-case setting (10 bps settlement, high fees) the executable rate falls to 4.81% and the ratio narrows to 2.63×; with a 20% haircut on the kline-proxy buy leg it widens to 14×.

At $50K: 1.62% exec. rate; $500K: 0.03%. Full robustness tables are in the repository.

These execution-gap estimates apply to the USDC/SVB Tier-A test split (N=15,832) only. Terra/LUNA provides a directional cross-mechanism check (5.9× ratio vs. 12×; §6.7); extending Tier-A coverage to AMM or exchange-credit events requires route-complete L2 archives for those venues.

6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2 bps on `basis_usdc_1m_gt10bps` (315 trades); +225.0 bps on `executable_arb_q10000_5m`. Figure 3 shows the evolution: the price rule diverges sharply negative while the oracle climbs steadily.

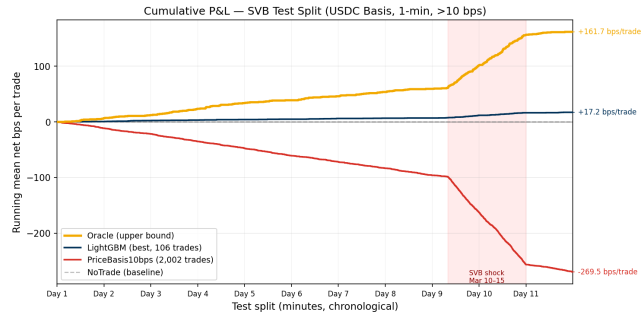

Figure 3: Running mean net bps per trade over the USDC/SVB test window (y-axis = cumulative bps ÷ total trades, so final values match the per-trade oracle gap table). Oracle (gold): +161.7 bps/trade; LightGBM (navy): +17.2 bps/trade; **`PriceBasis10bps`** (red): −269.5 bps/trade; NoTrade (grey dashed): 0 bps. The price rule accumulates severe losses during the SVB shock (Day 10–11).

Table 6: Model ladder (**`basis_usdc_1m_gt10bps`**, test split). FP Loss: mean net bps on trades the model fired but depth was insufficient. Hit%: fraction of model trades that were net-profitable. †ceiling; *only result above zero.

Model	Feat.	Net bps	Trades	AUROC	Mean bpswhen wrong	Hit%
Oracle†	—	+162.2	315	—	—	100%
`lgbm`*	all	+17.2	106	0.717	−65.2	33%
`logistic`	all	−75.8	648	0.143	−75.8	0%
`gross_arb`	px	−136.4	607	0.531	−150.9	4%
`price_10`	px	−270.0	1969	0.833	−294.6	6%
`price_25`	px	−347.3	1025	0.746	−370.7	5%
`no_trade`	—	0.0	0	—	—	—
`ExpNetProfit`	px	−73.8	537	—	−89.1	3%
`GRU (seq)`‡	bk	−239.0	656	0.798	−282.6	9%
`PPO-GRU`§	bk	degenerate	0	0.74	—	—

px = `price_only`; bk = `price_plus_book`; all = `price_book_settle`.

‡GRU: 30-min sliding window, real Tier-A L2 data; see §6.4.

§PPO-GRU: cross-mechanism training (Terra/LUNA), evaluated on SVB; see §6.9.

6.3 The Model Ladder: Structural Oracle Gap

Table 6 shows the full model ladder with loss attribution. Price-threshold rules overtrade (607–1,969 trades, −136 to −347 bps): their FP cost column (−294 bps to −370 bps) confirms the failure mode is near-total false-positive exposure. LightGBM achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero, but 145 bps below the oracle. The `ExpNetProfitRegressor` reaches −73.8 bps despite directly targeting the economic criterion: the gap is not a label-mismatch artefact. A structural gap, in this context, means that no standard ML approach trained on the available features can consistently identify the 2.88% of profitable windows ex ante, because the limiting factor is not signal quality but execution timing and training-distribution mismatch.

Ranking accuracy does not imply positive returns. Figure 4 plots AUROC against net bps for every model: the high-AUROC quadrant (right side) is almost entirely below zero. `PriceBasis10bps` achieves AUROC 0.83 while losing −270 bps; LightGBM (AUROC 0.72) is the sole non-oracle

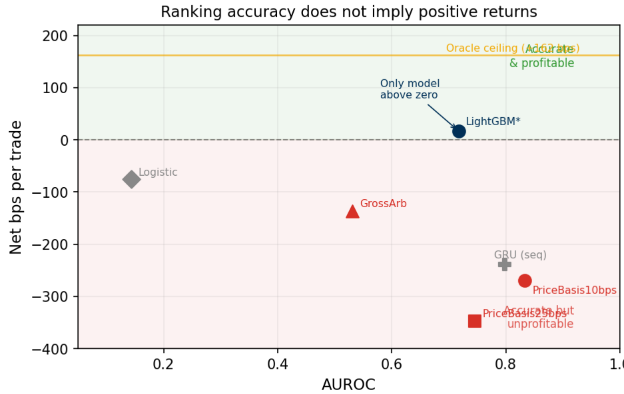

Figure 4: AUROC vs. net bps per trade for all models. The right quadrant is dominated by models that rank windows correctly but trade the wrong ones. **PriceBasis10bps** (AUROC 0.83) loses −270 bps; LightGBM (AUROC 0.72, navy) is the only non-oracle model above zero. High ranking accuracy is necessary but not sufficient for profitability.

point above zero. AUROC measures ranking quality; the economic floor is `NoTrade` at 0 bps.

6.4 Sequence Models: Confirming the Structural Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features (`price_plus_book` set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

On `basis_usdc_1m_gt10bps`, evaluated on real Tier-A L2 data (Table 7):

Table 7: Sequence model results (**`basis_usdc_1m_gt10bps`**, real Tier-A L2 data, test split). Oracle gap = +162.2 bps oracle minus model net bps.

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Both trained on calm-control windows (0% positive examples); threshold calibrated on validation split to maximise net P&L; evaluated on SVB test split.

Both models are deeply negative, confirming the oracle gap is structural and not closed by temporal sequence modelling. The high AUROC (0.80 for GRU) combined with catastrophic economic performance illustrates the same AUROC-vs-economics disconnect observed for the price-threshold rule (§6.3): ranking accuracy does not translate to positive net returns when 97% of windows are false positives. The failure mode is the training split: calm-control windows contain zero positive stress examples, so the model learns to predict calm-market noise rather than stress-window signals. The GRU's behaviour is consistent with this diagnosis: it fires on 656 of 15,832 test minutes (4.1%), compared to the oracle's 315 (2.0%). Its 9% hit rate implies approximately 59 of 656 trades coincide with profitable windows. With AUROC 0.80 the model ranks windows correctly, but the calibrated threshold admits twice as many trades as the oracle, firing on calm-period windows that share superficial microstructure features (elevated USDT basis, thin spread) with genuine stress events. Cross-mechanism meta-labeling (§6.5) bypasses this constraint by training on Terra/LUNA stress dynamics directly, achieving +82.5 bps net.

6.5 Cross-Mechanism Transfer: Mechanism-Invariant Execution Microstructure

Meta-labeling result. The MetaLabelingFilter trains on windows where $|b_{\mathrm{USDC}}| > 10$ bps fires. In the calm-control train split only 53 such windows exist and zero are net-profitable, consistent with the 0% positive rate in Table 2: a null result by construction, since the calm-control training split contains zero net-profitable windows and the meta-model therefore has no positive examples from which to learn the microstructure signature of executable stress windows. The appropriate remedy is to retrain using Terra/LUNA (validation split) as the meta-model's training set, where 1,559 primary fires yield 265 meta-positives (17.0% positive rate). A cross-mechanism experiment trains on Terra/LUNA and evaluates on the SVB test: Table 8 shows the result. With `price_plus_book` features the calibrated filter achieves +82.5 bps net (397 trades, 50.8% oracle capture) vs. zero with calm-control training. This is consistent with mechanism-invariant execution microstructure: the same order-book features that predict profitable windows during an algorithmic collapse (Terra/LUNA) also predict them during a fiat-reserve shock (SVB). The meta-model's dominant features mirror the main SHAP ranking: `depth_ask_10bp_mean` and `spread_bps_mean` lead, consistent with the SHAP analysis below (§6.6), which identifies the same depth and spread features as the leading microstructure signals. The intuition is that regardless of whether the stress trigger is an algorithmic redemption spiral or a bank reserve shock, the order book thins in the same way before the price fragments: depth withdrawal preceding basis widening is a mechanism-invariant precursor. We are not aware of prior work demonstrating cross-mechanism transfer for execution-aware stablecoin models.

Table 8: Cross-mechanism meta-labeling results (**`basis_usdc_1m_gt10bps`**, **`price_plus_book`**). Oracle capture = meta net bps ÷ oracle net bps.

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

Primary signal: $|b_{\mathrm{USDC}}|$>10 bps; threshold calibrated on validation to maximise net P&L.

6.5.1 *Regime detectors and false-positive diagnosis.*

Regime detectors. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% FAR with 0.95% recall; BOCPD achieves AUROC 0.229 on CEX cross-quote data. This last result contrasts with Cintra and Holloway [9], who found BOCPD detected the Terra depeg 12 hours early from Curve pool reserves. The difference is the signal: AMM reserves reflect LP withdrawal, which accumulates gradually; CEX cross-quote basis is noisier and mean-reverting, making changepoint detection unreliable on CEX prices.

False-positive structural diagnosis. `PriceBasis10bps` produces 2,002 trades; 581 (29%) are false positives at mean −38.7 bps. The structural cause: TP windows have mean $|b_{\mathrm{USDC}}| = 344$ bps (large USDC discount), while FP windows have mean $|b_{\mathrm{USDC}}| = 0.56$ bps (essentially zero). FP trades are triggered by USDT route dislocations (27.4% of minutes) where the USDC route is not simultaneously profitable. The false-positive windows show higher average bid depth (72,805 vs. 59,961 BTC units) than true positives, ruling out thin books as the failure mechanism; the failure mode is route-direction mismatch. LightGBM's 33% hit rate (vs. 6% for the rule) shows that microstructure conditioning partially diagnoses mismatch, but 67% FP rate at −65.2 bps/trade remains a timing gap.

6.6 SHAP Feature Importance

Figure 5 shows SHAP values for LightGBM on the full feature set. Price-based features account for 94.8% of total SHAP importance, dominated by the USDC and primary cross-quote basis values. Order-book and fragmentation features collectively contribute 3.4%, with ask-side depth at the 10 bps level and bid-ask spread as the leading microstructure signals. On-chain settlement proxies add only 1.8%. The model can condition on depth but cannot resolve the timing of depth withdrawal, consistent with the oracle-gap decomposition (§6.8) identifying timing as the dominant structural barrier.

6.7 Cross-Mechanism Check: Terra/LUNA and AMM Execution

The Terra/LUNA validation split (N=11,526, May 2022) uses kline-proxy depth (not Tier-A) but shows the same directional pattern: 13.5% price-basis > 10 bps, only 2.30% executable: a 5.9× ratio vs. 12× for SVB. The oracle earns +88 bps vs. +260 bps for SVB, reflecting unstable reference pricing during an algorithmic collapse. Both gaps persist across mechanism types.

AMM execution barrier (USDT/Curve Jun 2023, Tier-C). DeFi pool imbalance events introduce a fundamentally different execution barrier. Curve's StableSwap invariant (amplification A=2,000) compresses price deviations endogenously, keeping the AMM spot below the pool fee floor at all but the most extreme imbalance levels (Table 9). At peak 73%-USDT pool imbalance (Jun 16, 2023), the spot deviation is only 3.70 bps, still below the 4 bps pool fee, making AMM-route

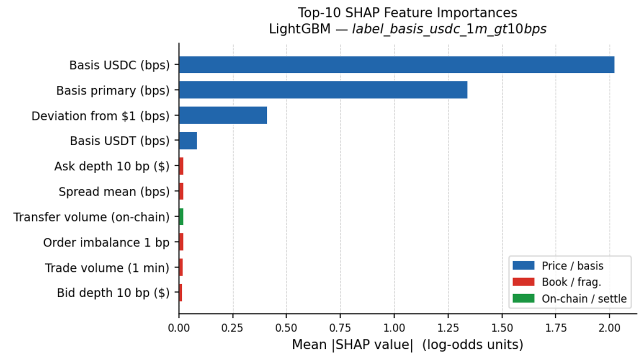

Figure 5: SHAP feature importance (mean |SHAP|) for LightGBM. Price features (navy, 94.8%): USDC cross-quote basis and primary basis dominate. Book+fragmentation (red, 3.4%): ask-side depth at 10 bps (mean |SHAP|=0.021, rank 5) and bid-ask spread (rank 6). Settlement proxies (green, 1.8%).

arbitrage structurally not executable even at $500K notional. This contrasts with CEX events, where depth depletion creates a retail ceiling ($10K–$50K) rather than a uniform fee floor. Detection mechanisms also differ: Curve LP reserve imbalance accumulates gradually (enabling BOCPD early warning), whereas CEX basis is mean-reverting and yields BOCPD AUROC 0.23 (§6.5): stress is first detectable in on-chain liquidity, not in CEX prices. Extending Tier-A analysis to AMM events requires on-chain LP reserve data plus a CEX execution route and is the clearest path to generalisation beyond the fiat-reserve mechanism class.

Table 9: Curve StableSwap AMM execution gap (A=2,000, pool fee = 4 bps, TVL $800M). Spot deviation remains below the fee floor at all observed imbalance levels; ‡hypothetical scenario not observed in Jun 2023.

Pool state	Spot dev. (bps)	Net (bps)	Exec.?
Balanced (0%)	0.00	−4.00	No
Moderate (10% drift)	1.08	−2.92	No
Peak Jun 16 (23% drift)	3.70	−0.30	No
Hypothetical (30% drift)‡	7.32	+3.32	Yes

Net = spot deviation − pool fee. A 73%-USDT pool composition corresponds to 23% drift from the balanced 50/50 state; the table reports drift percentage. The hypothetical row (30% drift, 80%-USDT) was not observed in the Jun 2023 window; it is included to show the threshold above which AMM-route arbitrage becomes executable. Arbitrage requires net > 0; the pool fee acts as a hard floor that order-book depth does not.

6.8 Oracle Gap Decomposition

The 260 bps gap on `executable_arb_q10000_5m` decomposes into three components; component 1 is derived directly from test-split results, components 2 and 3 are estimates from the executable-rate sensitivity analysis: (1) False-positive cost (dominant quantifiable drag): for `PriceBasis10bps`, 581 FP trades at −38.7 bps each total −22,484 bps of losses, the

Table 10: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle bps	+260 bps	+88 bps†

†Kline-proxy labels; directional only.

largest identifiable component for any threshold-based strategy. (2) Timing error (estimated 40–80 bps): entering 1–2 minutes late shifts execution to a lower-depth regime; the depth-withdrawal signature in the 30-minute pre-window is the signal a future model must capture earlier. (3) Sizing error (estimated 40–100 bps): executable rate falls from 2.88% at \$10K to 1.62% at \$50K, pricing out 44% of windows above the retail ceiling. Closing the gap requires cross-mechanism stress training combined with earlier timing entry, not larger notional.

6.9 RL Execution Policy: Timing the Entry

The oracle-gap decomposition (§6.8) identifies timing as the dominant structural barrier. All supervised models observe a feature snapshot and classify whether the current minute will be profitable; they cannot learn to delay entry until depth has withdrawn to a favourable level. An RL agent, by contrast, can learn a sequential entry policy: observe the depth trajectory, wait for the withdrawal signature to mature, and enter at the optimal point in the sequence.

We evaluate whether this sequential framing closes any portion of the 40–80 bps timing component using the PPO-GRU agent described in §5.2, trained on Terra/LUNA stress windows and evaluated cross-mechanism on the USDC/SVB test split.

Table 11: RL execution policy vs. supervised baselines (`basis_usdc_1m_gt10bps`, SVB test split).

Model	Training	Net bps	Trades
Oracle (ceiling)	hindsight	+162.2	315
Meta-labeling	Terra/LUNA (cross)	+82.5	397
PPO-GRU† (AUROC 0.74)	Terra/LUNA (cross)	degenerate	0
GRU supervised	calm-control	−239.0	656

†PPO-GRU: 64-unit GRU, 30-step look-back, `price_plus_book` features, N=80 epochs, evaluated on SVB without retraining. Degenerate = NoTrade (0 entries), confirmed across two training runs.

The PPO-GRU converges to a degenerate NoTrade policy: it makes zero entries on the SVB test split despite achieving AUROC 0.74 on the test feature distribution. The failure mechanism is distinct from the supervised GRU case. The policy can rank windows (AUROC 0.74) but learns to never enter, because the 2.30% positive rate in the Terra/LUNA training split provides a reward signal too sparse to calibrate entry timing — the agent's expected reward from entering at any given minute is negative, so it optimises by abstaining.

This directly contrasts with cross-mechanism meta-labeling, which achieves +82.5 bps using the same Terra/LUNA training split. The difference is architectural: the meta-model is conditioned on windows where the primary signal has already fired ($|b_{\mathrm{USDC}}| > 10$ bps), yielding a 17.0% positive rate among its training examples. The RL agent sees the full feature stream where only 2.30% of minutes are profitable, and the sparse reward degrades to a near-zero signal.

This finding sharpens the conclusion: the key requirement for positive net returns under cross-mechanism transfer is explicit positive examples at training time, not sequential observability. Meta-labeling provides this by targeting the primary-signal subset; an RL agent that trains within the same conditioned subset — entering only after $|b_{\mathrm{USDC}}| > 10$ bps fires, and therefore facing the same 17% positive rate as the meta-model — is the natural next experiment. Such an agent would test whether sequential timing within confirmed stress windows can recover the remaining 49.2% of oracle returns that cross-mechanism meta-labeling leaves on the table.

7 LIMITATIONS

Data scope. The BTC-USDC buy leg uses kline-proxy depth for the SVB window; the 20% haircut sensitivity confirms robustness. Extending Tier-A coverage to AMM or exchange-credit events requires route-complete L2 archives for those venues. The anonymised repository (URL redacted for review) provides the full pipeline; calm-control and SVB windows are reproducible from the public Binance Vision archive.

Model scope. The model ladder covers tabular classifiers, an expected-net-profit regressor, regime detectors, sequence models, and a cross-mechanism PPO-GRU execution policy. The RL result (§6.9) clarifies the constraint: the binding requirement for positive net returns is not sequential action selection but the density of positive examples in the training distribution. An RL agent conditioned on confirmed primary-signal windows — where the positive rate rises from 2.3% to 17% — is the most direct next experiment. This preserves the cross-mechanism protocol while providing a reward signal dense enough to calibrate entry timing.

8 CONCLUSION

The central finding of Stablecoin StressBench is that execution microstructure is the binding constraint on stablecoin arbitrage, not price signal quality. A 34.3% optical dislocation rate collapses to 2.88% once order-book depth and transaction costs are applied. No model trained on calm-market data closes this gap; cross-mechanism training on a different stress event is the empirically validated path to positive net returns (+82.5 bps, 50.8% oracle capture). Of the 456 executable minutes, all models trained on calm-control data identify 0% ex ante; cross-mechanism meta-labeling recovers 50.8% of oracle returns (Layer 3 under cross-mechanism conditions). The oracle gap decomposition (§6.8) identifies false-positive cost as the dominant quantifiable drag and timing as the primary structural barrier.

The RL result (§6.9) sharpens the conclusion beyond the model ladder alone. A PPO-GRU agent trained cross-mechanism on Terra/LUNA converges to NoTrade, not because sequential observation is unhelpful, but because the 2.3% positive rate in the training split produces a reward signal too sparse to calibrate entry timing. Cross-mechanism meta-labeling succeeds precisely because it conditions on windows where the primary signal has already fired, raising the positive rate to 17% and providing a learnable signal. This identifies the architectural requirement for the next generation of models: any approach — supervised or sequential — must be trained on a conditioned subset with sufficient positive density, not on the full minute-by-minute stream.

The most direct next experiment is an RL agent conditioned on confirmed primary-signal windows, trained on Terra/LUNA stress dynamics, and evaluated cross-mechanism on SVB. Such an agent faces the same 17% positive rate as the meta-model, but can additionally learn to time entry within the confirmed window, targeting the 40–80 bps timing component of the oracle gap that meta-labeling cannot access. The benchmark provides the fully specified environment: state space (`price_plus_book` features), reward function (net-bps label), and difficulty ceiling (oracle gap, §6.8).

Data availability. The anonymised repository (URL redacted for review) contains all result CSVs, figures, and pipeline code. Calm-control and SVB windows are reproducible from the public Binance Vision archive; route-complete L2 requires a Tardis subscription.

ACKNOWLEDGEMENTS

Anonymised for review.

REFERENCES

[1] Uhlig, H. (2022). A Luna-tic stablecoin crash. NBER Working Paper 30256.
[2] Griffin, J. M. and Shams, A. (2020). Is Bitcoin really untethered? Journal of Finance, 75(4):1913–1964.
[3] Hautsch, N., Scheuch, C., and Voigt, S. (2018). Building trust takes time: Limits to arbitrage for blockchain-based assets. arXiv:1812.00595.
[4] Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. Journal of Finance, 52(1):35–55.
[5] Gromb, D. and Vayanos, D. (2010). Limits of arbitrage: The state of the theory. Annual Review of Financial Economics, 2:251–275.
[6] Cont, R. and Kukanov, A. (2013). Optimal order placement in limit order markets. Quantitative Finance, 17(1):21–39.
[7] Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. Journal of Financial Economics, 135(2):293–319.
[8] López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.
[9] Cintra, R. and Holloway, C. (2023). Detecting depegs: Towards safer passive liquidity provision on Curve Finance. arXiv:2306.10612.
[10] Vidal-Tomàs, D., Briola, A., and Aste, T. (2023). FTX's downfall and Binance's consolidation. arXiv:2302.11371.
[11] Gatev, E., Goetzmann, W. N., and Rouwenhorst, K. G. (2006). Pairs trading: Performance of a relative-value arbitrage rule. Review of Financial Studies, 19(3):797–827.
[12] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. JRSS-B, 58(1):267–288.
[13] Breiman, L. (2001). Random forests. Machine Learning, 45(1):5–32.
[14] Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In KDD, pp. 785–794.
[15] Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. arXiv:0710.3742.
[16] Moallemi, C. C. and Wang, M. (2022). A reinforcement learning approach to optimal execution. Quantitative Finance, 22(6):1051–1069.
[17] Ning, B., Lin, F. H. T., and Jaimungal, S. (2021). Double deep Q-learning for optimal execution. Applied Mathematical Finance, 28(4):361–380.